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Operationalizing AI in Primary English Classrooms: A Thematic Qualitative Synthesis of AI PengTalk Application*

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This study investigates the pedagogical mechanisms underlying the integration of 'AI PengTalk,' a government-initiated AI speaking practice system, into South Korean primary English education. To address the gap in understanding its classroom operationalization, a thematic qualitative synthesis of 36 empirical studies was conducted. Instructional models, learner affordances and perceptions, and teacher and peer mediation strategies in response to technological and assessment constraints were systematically analyzed. The findings reveal that AI PengTalk is most prominently operationalized within Flipped Learning, Spaced Practice, and Differentiated Remedial Instruction models. Pedagogically, the tool creates a 'psychological safety zone' that significantly lowers the affective filter for lower-proficiency learners, fostering a sense of social presence and motivation through gamification. However, it simultaneously exhibits a 'ceiling effect' and a fading novelty for advanced students due to the lack of sophisticated corrective feedback. Furthermore, technical and assessment constraints—such as speech recognition errors and rigid grading algorithms—necessitate that teachers evolve into active pedagogical mediators who manage learner frustration, supported by peer collaborative scaffolding. The study concludes that successful AI integration does not replace human instruction but requires a restructuring of the teacher's role to combine high-tech data utilization with high-touch emotional support.

Key Words: AI chatbot, AI PengTalk, AI 펭톡, operationalization, thematic qualitative synthesis

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I. Introduction

The integration of Artificial Intelligence (AI) technologies into English language education has become increasingly prevalent globally (H. Lee & Lee, 2024; Zhang & Zou, 2022). Particularly in English as a Foreign Language (EFL) contexts where authentic input and output opportunities are scarce, AI-powered tools such as conversational chatbots have emerged as innovative solutions to overcome temporal and spatial constraints (Yang et al., 2022). By utilizing Natural Language Processing (NLP) and Automatic Speech Recognition (ASR), these AI tools serve as personalized interactive partners, providing learners with individualized feedback and expanded communicative practices (Jeong & Choi, 2021; B. O. Kang et al., 2024). In line with this global trend, AI PengTalk—a government-initiated, AI-based English speaking practice system co-developed by the Ministry of Education and EBS and launched in 2021—distinguishes itself from generic chatbots through its close alignment with the national curriculum. Built upon the 2015 and 2022 Revised National English Curricula, it allows students to practice curriculum-aligned phrases and receive real-time feedback on their pronunciation. Furthermore, by incorporating gamification elements featuring the popular character 'Pengsoo,' the system aims to lower the affective filter and enhance learner immersion.

However, despite the proliferation of empirical studies following its widespread adoption, the current research landscape remains inconsistent. Existing studies have largely focused on proving whether the tool is 'effective' in isolated contexts—such as checking improvements in pronunciation or vocabulary—or have been limited to short-term perception surveys. While these studies confirm the tool's general utility, they often overlook its 'operationalization'—the practical implementation and pedagogical integration of the technology into classroom instructional design to reshape learning processes. As the initial novelty of AI fades, it becomes evident that the technology does not function in a vacuum; its success heavily relies on how it is structured within a lesson and how teachers guide the learning process.

Moreover, conflicting findings regarding learner perceptions suggest that the tool's affordances are not universal but are moderated by learner characteristics such as proficiency and grade level. Additionally, while technological potential is high, practical constraints such as speech recognition errors and the 'novelty effect' necessitate active teacher mediation, yet this aspect remains under-synthesized in current literature.

Therefore, this study aims to fill these gaps by conducting a thematic qualitative

synthesis of empirical studies related to AI PengTalk. Unlike previous reviews (e.g., Huang et al., 2022; J. Y. Lee & Hwang, 2022) that broadly examined generic chatbots (e.g., Alexa) or solely synthesized quantitative efficacy data (e.g., test scores), this study specifically investigates the operational patterns and pedagogical dynamics of a customized AI tool within the Korean primary English context. To achieve this, the current research comprehensively synthesizes qualitative evidence regarding how the AI tool is integrated into diverse instructional models, how its pedagogical affordances are perceived by different learners, and what mediation strategies teachers employ to overcome practical constraints. By doing so, this research seeks to derive theoretical and practical implications for establishing sustainable models of AI-human collaboration in future EFL classrooms.

II. Literature Review

1. Theoretical Background

The integration of AI chatbots into English as a Foreign Language (EFL) education is theoretically grounded in three foundational perspectives: cognitive, affective, and sociocultural. First, from a cognitive perspective, AI PengTalk serves as an effective tool for facilitating 'pushed output' in EFL environments where authentic speaking opportunities are scarce. Based on Swain's (1985) Output Hypothesis, teachers frequently utilize the AI as a pre-class flipped learning assignment, compelling learners to cognitively rehearse vocabulary and syntax before engaging in classroom interactions (Cha, 2023; Jeon, 2023; Y. Lee & Lee, 2024). Furthermore, recent studies suggest that the cognitive benefits of such AI-assisted learning are maximized when integrated with spaced practice (S. H. Kang et al., 2014; Nakata, 2015); systematically scheduling repetitive speaking tasks at expanding intervals effectively counters memory decay and promotes long-term retention (M. Kim et al., 2023; Kwon & Shim, 2023).

Second, from an affective perspective, interacting with an AI chatbot provides a non-judgmental environment that significantly lowers learners' speaking anxiety and affective filter (Krashen, 1982). This psychological safety encourages students to participate more actively without the fear of making mistakes in front of peers, often driven by gamified elements such as digital rewards (Heo, 2024; S. Lee, 2019; Woo & Kim, 2023). This reduction in speaking anxiety is profoundly amplified when learners

perceive the AI as a social actor rather than a mere machine, developing a sense of social presence and perceived homophily with the AI character (Nass & Moon, 2000; T. Yoon, 2024). However, literature also warns of a fading novelty effect, indicating that without challenging tasks, initial interest may drop rapidly, particularly among advanced learners who find mechanical repetitions unchallenging (W. Choi, 2021; Jeong et al., 2021; Seong & Lee, 2021).

Third, from a sociocultural perspective, the introduction of AI into the classroom is deeply connected to Vygotsky's (1978) Zone of Proximal Development, highlighting the importance of the TPACK (Technological Pedagogical Content Knowledge) framework (Mishra & Koehler, 2006). Specifically, because current ASR algorithms still struggle to distinguish specific phonemes and exhibit rigid grading patterns compared to human raters (Derwing & Munro, 2015; M. Park & Lee, 2022; Yoo & Ahn, 2024), unmediated AI use can cause learner frustration. This necessitates hybrid scaffolding, where teachers actively combine AI's digital affordances with high-touch human interventions, such as providing nuanced corrective feedback and facilitating peer collaboration (Oh et al., 2022; Rim & Han, 2024; Um et al., 2023; Yeon, 2023). Consequently, this study explicitly adopts the TPACK framework as its primary theoretical lens to explore how teachers orchestrate these interactions.

2. Application of AI Chatbots in EFL Contexts

A growing body of global EFL research has demonstrated the efficacy of AI chatbots in enhancing language acquisition by providing interactive communication opportunities (e.g., Chen Hsieh et al., 2017; Huang et al., 2022; H. Lee & Lee, 2024). Within this global trend, various commercial virtual assistants, such as Alexa, have been successfully integrated into language learning environments, primarily focusing on improving the communicative competence of adult or university learners (N. Kim, 2017; H. Lee & Lee, 2025).

Aligning with these active global applications, researchers in the domestic context of South Korea have further expanded the scope by exploring a wide spectrum of AI chatbots specifically tailored to overcome the lack of authentic English exposure for young learners (Y. B. Yoon & Park, 2020). Initial domestic studies applied commercial smart speakers—such as Google Assistant, Amazon Alexa, Mitsuku, Cleverbot, and Clova's Monica—to English classrooms (H. Kim et al., 2019). These studies revealed that interacting with such devices provided learners with novel conversational

opportunities and significantly lowered their foreign language anxiety (e.g., Hyun & Im, 2019; J. H. Lee et al., 2019; Shin, 2019). However, researchers soon identified critical limitations in using these non-pedagogical chatbots for young learners, noting their difficulty in recognizing Korean-accented English, the high linguistic difficulty of their rapid responses, and a lack of curriculum alignment (W. Choi, 2021; Jeong & Choi, 2021; M. Park & Lee, 2022; Shin, 2019).

To address these constraints, subsequent domestic research pivoted towards developing and applying custom-built, educational chatbots tailored specifically for young learners. Researchers utilized platforms like Google Dialogflow to design task-based chatbots (Han, 2021; Hwang & Lee, 2021; Sung, 2019), while others applied specifically developed educational bots like 'Ellie' (Yang et al., 2019). Studies utilizing these bespoke chatbots demonstrated significant improvements in elementary students' speaking volume, fluency, and affective domains, proving the potential of pedagogically designed AI as a powerful interactive partner (Chu & Min, 2019; Kim & Lee, 2020; S. Lee, 2019; Shim et al., 2020).

3. Trends and Research Gaps in AI PengTalk Studies

Since its nationwide deployment in 2021, a proliferation of empirical studies has investigated the educational utility of AI PengTalk in South Korean primary English classrooms. Most existing research has predominantly focused on quantitatively verifying the tool's effectiveness in isolated contexts, such as measuring short-term improvements in pronunciation, vocabulary, and speaking fluency (Cha, 2023; Jang & Kim, 2022; H. Kim et al., 2021; K. Lee, 2022; Oh & Back, 2022; Seong, 2022; Woo & Kim, 2023; Yoo & Ahn, 2024; T. Yoon, 2022). Additionally, numerous studies have explored the affective benefits of the application, reporting that its gamification elements and non-judgmental environment successfully reduce learners' speaking anxiety and foster positive attitudes toward English learning (W. Choi, 2021; Heo, 2024; Jeon, 2023; Ki, 2021; S. Kim, 2024; Seong & Lee, 2021; S. Yoo, 2024; T. Yoon & Lee, 2021).

However, despite these growing trends, the current research landscape remains fragmented and exhibits critical gaps. First, while previous studies confirm the tool's general utility, they often overlook its 'operationalization'—how the technology is practically implemented and pedagogically integrated into diverse instructional models within the authentic classroom ecosystem. Second, conflicting findings regarding learner perceptions suggest that the tool's affordances are not universal but are heavily

moderated by learner characteristics such as proficiency levels (S. Kim, 2024; Oh & Back, 2022). There is a distinct gap in understanding the paradoxical affective outcomes, such as why the initial novelty effect tends to fade quickly among advanced learners who find the mechanical feedback unchallenging (W. Choi, 2021; Jeong et al., 2021; Seong & Lee, 2021). Finally, while practical constraints like ASR errors and algorithmic limitations are frequently reported (M. Park & Lee, 2022; Yeon, 2023; Yoo & Ahn, 2024), the specific 'hybrid scaffolding' strategies that teachers and peers actively employ to mediate these technical challenges remain under-synthesized in current literature (Rim & Han, 2024; Um et al., 2023). Therefore, a comprehensive qualitative synthesis is urgently needed to uncover the underlying pedagogical mechanisms of AI PengTalk beyond simple efficacy tests.

III. Current Study

To address the gaps identified in the literature, the current study aims to bridge these inconsistencies by synthesizing qualitative evidence from primary studies. Rather than merely re-confirming effectiveness, this study focuses on uncovering the pedagogical mechanisms of how AI PengTalk is operationalized, perceived, and mediated in authentic classroom settings. Specifically, this synthesis moves beyond a binary assessment of success or failure to categorize the dominant instructional models, analyze how learner variables moderate the tool's affordances, and identify the strategies teachers employ to mediate technological constraints. By doing so, this research seeks to derive theoretical and practical implications for sustainable AI integration in EFL education. In line with these objectives, this study addresses the following three research questions:

RQ#1 (Operationalization Models): How is AI PengTalk operationalized and integrated into primary English curriculums?

RQ#2 (Affordances & Perceptions): What are the pedagogical affordances and limitations perceived by learners and teachers across different proficiency and grade levels?

RQ#3 (Constraints & Mediation): What are the technological and assessment constraints encountered in classrooms, and how do teachers and peers mediate these challenges?

IV. Methods

1. Research Design

This study employs a thematic qualitative synthesis design to investigate the pedagogical mechanisms of AI PengTalk in South Korean primary English classrooms. Unlike previous quantitative meta-analyses that focused primarily on efficacy metrics (e.g., test scores), this study aims to synthesize qualitative evidence to understand how the tool is operationalized within the curriculum, how learners and teachers perceive its affordances, and how contextual constraints are mediated. By integrating narrative data from observation logs, in-depth interviews, and open-ended survey responses across the identified empirical studies, this research seeks to reconstruct the lived experiences and instructional dynamics of AI-based English education.

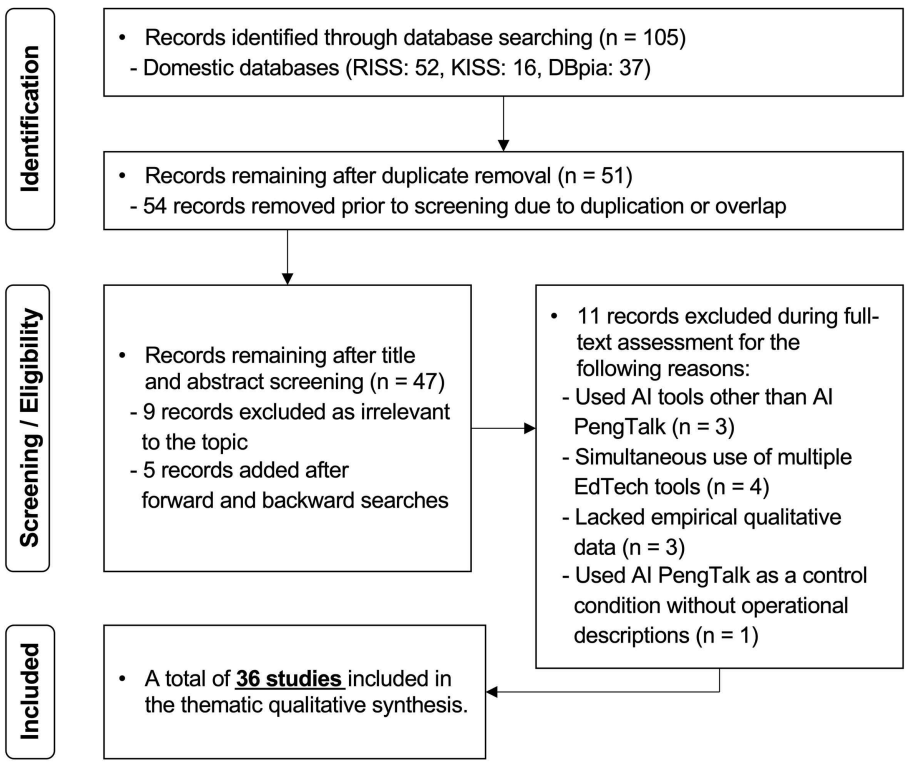
2. Literature Search and Selection Process

A systematic literature search was conducted using major domestic academic databases, including RISS, KISS, and DBpia. To ensure data relevance and specificity, the search was restricted to the keywords ‘AI 펭톡,’ ‘펭톡,’ ‘AI PengTalk,’ and ‘PengTalk.’ The search period extended from AI PengTalk’s 2021 public launch to the present.

Following an exhaustive search of research related to AI PengTalk in Korean elementary English education, the final selection was narrowed by applying four rigorous exclusion criteria to ensure alignment with the study’s analytical objectives. Specifically, studies were excluded if they (1) focused on general AI chatbots rather than AI PengTalk; (2) involved the simultaneous use of multiple EdTech tools (e.g., Metaverse), making it impossible to isolate the specific context of AI PengTalk usage; (3) lacked empirical data (e.g., simple program proposals or software reviews); or (4) used AI PengTalk solely as a control condition without detailed descriptions of its operationalization.

As shown in <Figure 1>, which illustrates the stages of the literature search and selection process, the initial search yielded a total of 105 records (RISS: 52, KISS: 16, DBpia: 37). After removing duplicates, 51 studies remained. A screening of titles and abstracts led to the exclusion of 9 studies deemed irrelevant, leaving 42 studies for full-text review. During this stage, an additional 5 studies were identified through forward and backward searches, resulting in a total of 47 studies for in-depth analysis. Upon reviewing these 47 studies, 11 were further excluded based on the pre-established criteria.

Specifically, three studies were excluded as they utilized AI tools other than AI PengTalk (Criterion 1), and four were excluded because they involved the simultaneous use of multiple EdTech tools, making it impossible to isolate the specific context of AI PengTalk usage (Criterion 2). Additionally, three studies—including review papers and technical reports on the software's capabilities—were removed due to a lack of empirical qualitative data (Criterion 3). Finally, one study was excluded for using AI PengTalk solely as a control condition without providing detailed operational descriptions (Criterion 4). Ultimately, 36 studies were selected as the final dataset for this qualitative synthesis.



<Figure1> PRISMA Flowchart for Literature Search and Selection Process

3. Data Analysis Strategy

Data extracted from the 36 selected studies were analyzed using Thematic Synthesis (Thomas & Harden, 2008). To ensure the transparency of the analysis, a comprehensive

three-stage coding procedure was applied in this study. The process began with extracting Initial Codes inductively from the raw data (Step 1). During this step, to enhance methodological rigor, both positive outcomes and negative cases (e.g., practical constraints) were intentionally extracted. These codes were then grouped into categorical Descriptive Themes (Step 2). Finally, through a deductive mapping process, these themes were synthesized into Analytical Themes that directly address the three RQs (Step 3).

To establish the reliability of the coding process, an initial codebook was developed by two researchers—including the author—who independently coded 10 cases, representing 28% of the total dataset. Discrepancies were thoroughly discussed and resolved through subsequent consensus meetings to finalize the analytical themes. Since the primary purpose of this thematic qualitative synthesis is to generate themes rather than establish causality, a formal quality appraisal process was omitted in accordance with Thomas and Harden (2008).

V. Results

Based on the thematic qualitative synthesis of the empirical studies, this section presents the findings organized by the three research questions, mapping the descriptive themes to analytical themes. To provide a comprehensive overview of the findings, <Table 1> summarizes the thematic synthesis results, illustrating the extracted themes along with the frequency of studies (n) corresponding to each analytical theme.

<Table 1> Thematic Qualitative Synthesis of AI PengTalk Application

Initial Codes (Step 1)	Descriptive Themes (Step 2)	Analytical Themes (Step 3)
• Home-based assignments • Pre-class learning • Utilizing 'Topic World' • In-class supplementary tool	• Connecting online-offline learning and pre-learning • Utilizing as a supplementary tool in regular classes	Blended & Flipped Learning (RQ#1) (n = 18)
• Phonics supplementation • Individualized pace • Basic literacy training	• Individualized pacing for underachievers • Phonics and reading guidance for slow learners	Differentiated & Remedial Instruction (RQ#1) (n = 12)
• Considering forgetting curve • Repetitive exposure • Cumulative review	• Repetitive scheduling at expanding intervals • Structured cognitive rehearsal	Spaced Practice (RQ#1) (n = 7)

• 1, 3, 8-day interval practice		
• Lowering affective filter • Non-judgmental environment • Reduced shyness • Digital rewards & character customization	• Eliminating peer pressure and reducing speaking anxiety • Inducing intrinsic/extrinsic motivation via digital rewards	Affective Filter Reduction & Gamification (RQ#2) (n = 25)
• Treating AI as a conversational partner • Perceived homophily • Emotional intimacy with Pengsoo	• Emotional connection with the AI character • Granting presence as a social interaction partner	Social Presence & Homophily (RQ#2) (n = 9)
• Boredom of high achievers • Avoidance of simple repetition • Limits of free conversation	• Lack of challenging dialogue for advanced students • Decline in initial interest over time • Simple evaluative feedback instead of corrective feedback	Fading Novelty & Ceiling Effect (RQ#2) (n = 14)
• Speech recognition errors • System instability • Inability to distinguish fricatives/plosives • Overly strict grading compared to native speakers	• Frequent ASR failures • Technical limits in identifying specific phonemes • Conservativeness and inconsistency in AI assessment algorithms	Technological & Assessment Constraints (RQ#3) (n = 20)
• Teacher's emotional support • Resolving device errors • Providing pronunciation corrections missed by AI • Combined reward system	• Resolving technical issues during class • Teacher interventions to supplement AI's mechanical feedback • Emotional support and instructional redesign	Teacher as a Multifaceted Mediator (RQ#3) (n = 22)
• Pairing high/low proficiency students • Assigning group goals • Providing peer pronunciation hints	• Pairing heterogeneous (high-low) groups • Collaborative group rewards to offset AI limitations	Peer Collaborative Mediation (RQ#3) (n = 11)

1. Operationalization Models (RQ#1)

The synthesis revealed three distinct operational models integrating AI PengTalk into primary English curriculums. Most prominently, teachers employed a blended and flipped learning approach (n = 18) where the AI's 'Topic World' function was assigned as pre-class home learning, allowing valuable classroom time to be reserved for interactive communicative activities (e.g., Cha, 2023; Jeon, 2023; Jeong, 2025; Y. Lee & Lee, 2024). Additionally, the AI served as a differentiated and remedial tool (n = 12) for slow learners, offering individualized pacing for phonics and basic literacy training to

prevent learning deficits (e.g., S. Choi & Yoon, 2024; W. Choi, 2024; Y. Choi, 2024; M. Kim, 2026; S. Park & Kang, 2025; H. Yoon & Yu, 2021). Finally, a spaced practice model (n = 7) was identified, wherein repetitive speaking tasks were scheduled at expanding intervals (e.g., 1, 3, and 8 days) to systematically counter the forgetting curve and consolidate learners' long-term memory (e.g., S. Choi & Yoon, 2024; M. Kim, 2026; M. Kim et al., 2023).

2. Affordances and Perceptions (RQ#2)

Learners and teachers perceived several pedagogical affordances alongside notable limitations. The primary psychological benefit was affective filter reduction and motivation through gamification (n = 25), as the AI provided a non-judgmental practice environment free from peer pressure (e.g., Y. Choi, 2024; Heo, 2024; Woo & Kim, 2023; S. Yoo, 2024), bolstered by digital rewards and character customization that induced intrinsic and extrinsic motivation (e.g., Y. Choi, 2024; Ki, 2021; S. Lee, 2019; Seong, 2022; T. Yoon & Lee, 2021). Moreover, students often treated the AI character as a social actor, developing a sense of social presence and perceived homophily (n = 9) that enhanced their communicative willingness (e.g., Hong et al., 2021; S. Kim, 2024; Yeon, 2023; T. Yoon, 2024). Despite these affordances, fading novelty and a ceiling effect (n = 14) were observed among advanced learners. Their initial interest faded over time as they found the pre-programmed dialogues unchallenging and noted a lack of sophisticated corrective scaffolding (e.g., C. Choi & Lee, 2023; S. Choi & Yoon, 2024; W. Choi, 2021; Jeong et al., 2021).

3. Constraints and Mediation (RQ#3)

The practical implementation of AI PengTalk was frequently hindered by technological and assessment constraints (n = 20). Studies consistently reported ASR failures, the AI's inability to distinguish specific phonemes (e.g., fricatives and plosives), and overly strict or inconsistent grading compared to human raters (e.g., C. Choi & Lee, 2023; Hong et al., 2021; M. Park, 2023; M. Park & Lee, 2022; Yoo & Ahn, 2024). To offset these limitations, the teacher as a multifaceted mediator (n = 22) played a crucial role, providing the emotional support and nuanced corrective feedback that the AI lacked while resolving in-class technical issues (e.g., S. Kim, 2024; Oh et al., 2022; Seong & Lee, 2021; Yeon, 2023). Furthermore, peer collaborative mediation (n = 11) was widely

employed; teachers paired high- and low-proficiency students and assigned group goals, leveraging human-to-human scaffolding to compensate for the AI's mechanical shortcomings (e.g., S. Kim, 2024; K. Lee, 2022; Oh & Back, 2022).

VI. Discussion

This study synthesized empirical evidence from 36 primary studies to analyze how AI PengTalk is operationalized within Korean primary English classrooms. The findings suggest that AI PengTalk functions as a 'mediated tool' that dynamically interacts with learners' cognitive and affective mechanisms, rather than merely serving as a functional substitute for human instruction. To establish a robust connection with theoretical frameworks, the synthesized findings are interpreted through three distinct lenses: cognitive mechanisms, affective dimensions, and sociocultural dynamics.

1. Cognitive Mechanisms: Output Hypothesis and Spaced Practice

The operationalization of AI PengTalk primarily through flipped learning and spaced practice empirically supports cognitive theories of language acquisition, most notably Swain's (1985) Output Hypothesis. In traditional EFL environments like South Korea, learners frequently suffer from a lack of authentic output opportunities. By assigning AI speaking tasks—such as the 'Topic World'—as pre-class activities, teachers enforce 'pushed output' (Y. Lee & Lee, 2024). This flipped approach compels learners to move from semantic processing to syntactic processing independently, forcing them to cognitively rehearse vocabulary and syntax before engaging in real-time classroom interactions (Cha, 2023; Jeon, 2023).

Furthermore, the integration of the spaced practice model demonstrates a highly structured cognitive approach to countering memory decay. By utilizing the AI's capability to schedule repetitive speaking tasks at expanding time intervals (e.g., 1, 3, and 8 days), learners can transfer newly acquired vocabulary and phrases into their long-term memory much more effectively than through massed practice (M. Kim et al., 2023). This indicates that the AI tool's true cognitive value lies not just in the content it delivers, but in how it systematically structures the timing and frequency of learners' cognitive rehearsal.

2. Affective Dimensions: The Paradox of AI Scaffolding

The perceived affordances of AI PengTalk reveal a dual pedagogical effect heavily moderated by learner proficiency, which this study defines as the 'Paradox of AI Scaffolding.' This paradox can be comprehensively understood through Krashen's (1982) Affective Filter Hypothesis and Vygotsky's (1978) Zone of Proximal Development (ZPD).

On one hand, for low-proficiency learners, the AI serves as a highly effective affective scaffold. It acts as a non-judgmental social actor that completely removes the social anxiety associated with human peer evaluation, thereby lowering the affective filter and fostering a 'psychological safety zone' where infinite retries are permitted (W. Choi, 2024; S. Park & Kang, 2025; Woo & Kim, 2023). This psychological safety is further strengthened when learners establish a sense of 'social presence' and 'perceived homophily' with the AI character, Pengsoo, treating the machine as an empathetic conversational partner (T. Yoon, 2024).

On the other hand, for high-proficiency learners, the tool exhibits a 'ceiling effect,' representing a failure in cognitive scaffolding within the ZPD. Advanced learners quickly realize that the AI's automated assessment provides simple 'evaluative feedback' (e.g., 'Excellent' or 'Try harder') rather than the nuanced 'corrective feedback' or meaning negotiation required for higher-level cognitive challenges (W. Choi, 2021; Jeong et al., 2021). Consequently, the initial novelty effect fades, and these learners perceive the tasks as mechanical and unchallenging. This paradox implies that while current AI excels at affective scaffolding for beginners, it has not yet reached the status of a More Knowledgeable Other (MKO) capable of providing the complex cognitive scaffold required for advanced learners.

3. Sociocultural Dynamics: TPACK and Hybrid Scaffolding

The practical constraints of AI PengTalk highlight that its classroom integration must be understood through the lens of Activity Theory (Engeström, 1987) and the TPACK (Technological Pedagogical Content Knowledge) framework (Mishra & Koehler, 2006). The introduction of AI into the classroom ecosystem inevitably creates systemic contradictions, particularly due to technological limits. Studies consistently report that the current ASR algorithms struggle to distinguish specific English phonemes—such as fricatives and plosives—and often exhibit overly conservative grading patterns compared to human raters (M. Park & Lee, 2022; Yoo & Ahn, 2024).

These mechanical flaws cause learner frustration and disrupt the continuous learning process (Yeon, 2023). To resolve these contradictions, unmediated AI use is insufficient; instead, teachers enact 'hybrid scaffolding' (Um et al., 2023). By restructuring their TPACK competencies, teachers step in as multifaceted mediators—resolving technical glitches, providing the specific pronunciation corrections the AI misses, and offering crucial emotional support (Oh et al., 2022; Seong & Lee, 2021). Additionally, teachers utilize peer collaborative mediation by pairing high- and low-proficiency students, leveraging human-to-human interaction to compensate for the AI's mechanical shortcomings (Rim & Han, 2024). Ultimately, successful AI integration does not replace human instruction but necessitates a sophisticated blend of high-tech digital affordances and high-touch human interventions.

VII. Conclusion

This thematic qualitative synthesis of 36 empirical studies provides a comprehensive understanding of how AI PengTalk is operationalized within South Korean primary English classrooms. The findings demonstrate that AI PengTalk functions most effectively not as an independent instructor, but as a cognitive rehearsal tool integrated into blended learning frameworks, such as flipped learning, spaced practice, and differentiated remedial instruction (Cha, 2023; M. Kim et al., 2023; Y. Lee & Lee, 2024). Furthermore, this study highlights the 'Paradox of AI Scaffolding.' While the AI platform successfully lowers the affective filter by creating a non-judgmental psychological safety zone and fostering motivation through gamification and social presence for low-proficiency learners (W. Choi, 2024; S. Park & Kang, 2025; T. Yoon, 2024), it simultaneously exhibits a ceiling effect for advanced learners due to its mechanical feedback and inability to handle complex communicative tasks (W. Choi, 2021). Consequently, sustainable AI integration requires the active presence of teachers who employ hybrid scaffolding—combining digital affordances with human empathy—to mediate technological and assessment constraints like speech recognition errors, supported by peer collaborative mediation (M. Park & Lee, 2022; Rim & Han, 2024; Yeon, 2023).

Despite the rigorous synthesis of qualitative data, several limitations must be acknowledged. First, a majority of the primary studies included in this review were short-term experiments lasting less than one semester. This duration makes it challenging to disentangle genuine, long-term language acquisition from the temporary novelty effect

commonly observed when new technologies are introduced (C. Choi & Lee, 2023; S. Choi & Yoon, 2024; W. Choi, 2021; W. Choi, 2024). Second, from a technological standpoint, the current AI system demonstrates notable weaknesses in accurately assessing suprasegmental features, such as intonation and rhythm, struggles to distinguish specific phonemes, and applies overly rigid grading algorithms, which restricts its utility for advanced learners (C. Choi & Lee, 2023; M. Park, 2023; M. Park & Lee, 2022; Yoo & Ahn, 2024). Methodologically, this synthesis focused exclusively on domestic databases and the South Korean EFL context, as AI PengTalk is a customized platform for the national curriculum. While this provides highly contextualized insights, it limits the direct global generalizability of the findings. Additionally, a formal Quality Appraisal (QA) of the primary studies was omitted, as the primary goal was generative thematic synthesis rather than establishing empirical causality (Thomas & Harden, 2008).

To address these limitations, future research should pursue several directions. Methodologically, subsequent studies should conduct a quantitative meta-analysis to statistically evaluate the overall effect sizes of AI PengTalk on learners' cognitive and affective domains. Furthermore, longitudinal studies spanning over an academic year are urgently needed to verify the sustainability of learner motivation and learning retention beyond the initial novelty period. Pedagogically, further investigation should focus on developing and refining 'Hybrid Scaffolding' models and peer collaboration strategies that seamlessly link AI-based individual practice outside the classroom with meaningful, human-to-human communicative activities within the classroom. Technologically, development efforts must prioritize enhancing the accuracy of ASR algorithms, particularly in evaluating prosodic features and providing pedagogically flexible corrective feedback, to overcome the current ceiling effect and adequately support learners across all proficiency levels.

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An asterisk (*) denotes studies included in this thematic qualitative synthesis.

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